CGPS variables potentially used to define the outcome

Compiled from `header_BL.xlsx`, `header_FU.xlsx` and the aggregate report of `cgps_stats.py`

This document lists every variable in the CGPS tables that could be used to define the outcome of the risk prediction model, or eligibility for it. The CT volumes are not listed. BL and FU denote the baseline and follow-up tables; a tick means that the variable exists in that table. The role column refers to the chosen design (Section 6): *outcome*, *eligibility*, *check* (sensitivity or noise check), *covariate*, or *not used*. Counts are aggregates from the report on 2 116 patients with both scans.

1 Segments

Segment-level variables are named `ct_cag_<segment>_<item>`. The 17 segments are listed below. Only the trunk segments of the LAD, CX and RCA enter the chosen design; the LM enters as an eligibility criterion for the LAD and CX.

Code	Segment	Vessel	Design role	Note
`lm`	Left main	LM	eligibility	Must be plaque-free for LAD and CX units
`plad`	Proximal LAD	LAD	outcome	Trunk; 446 of 1 003 plaque-free at BL
`mlad`	Mid LAD	LAD	outcome	Trunk
`dlad`	Distal LAD	LAD	outcome	Trunk
`d1`	First diagonal	LAD	not used	Side branch
`d2`	Second diagonal	LAD	not used	Side branch
`pcx`	Proximal circumflex	CX	outcome	Trunk
`mcx`	Mid circumflex	CX	outcome	Trunk
`dcx`	Distal circumflex	CX	outcome	Trunk
`om1`	First obtuse marginal	CX	not used	Side branch
`om2`	Second obtuse marginal	CX	not used	Side branch
`prca`	Proximal RCA	RCA	outcome	Trunk

Code	Segment	Vessel	Design role	Note
`mrca`	Mid RCA	RCA	outcome	Trunk; most often unevaluable (motion)
`drca`	Distal RCA	RCA	outcome	Trunk; ends at the PDA origin
`pda`	Posterior descending artery	RCA/CX	not used	Side branch; origin depends on dominance
`pla`	Posterolateral branch	RCA/CX	not used	Side branch; origin depends on dominance
`im`	Ramus intermedius	–	not used	Present in a minority; often coded unevaluable when absent

1.1 Per-segment items

Variable	BL	FU	Values	Role	Note
`ct_cag_<s>_stenosis`	✓	✓	Text: none, minimal, mild, moderate, severe, occlusion, `uneval_*`	outcome, eligibility	Decoded to grades 0–5; `uneval_*` treated as missing. Empty at BL for 1 094 patients (not read).
`ct_cag_<s>_plaque_type`	✓	✓	Text: calcified, mostly_calcified, mostly_noncalcified, noncalcified; empty when no plaque	check	Composition of the new plaque; plaque presence is taken from the stenosis grade.
`ct_cag_<s>_feat_spotcalc`	✓	✓	0/1	not used	High-risk feature; recorded only with plaque; mostly noise (opposite changes exceed new findings).
`ct_cag_<s>_feat_napring`	✓	✓	0/1	not used	High-risk feature; rare.
`ct_cag_<s>_feat_lap`	–	✓	0/1	not used	Low-attenuation plaque; 0 for every patient at FU.
`ct_cag_<s>_feat_posrem`	–	✓	0/1	not used	Positive remodelling; FU only, so no change can be computed.
`ct_cag_<s>_eval`	✓	–	Numeric code (0, 1, 5, 12–14)	check	Evaluability; meaning of the codes unknown, to be confirmed with the data manager.
`ct_cag_<s>_medis_egnet`	✓	–	Numeric	not used	Suitability for Medis plaque analysis.

Variable	BL	FU	Values	Role	Note
ct_cag_<s>_medis_finished	✓	–	Numeric	check	Medis analysis done (≈200 patients); exports may contain plaque positions along the centreline.

1.2 Stenosis grading

The grade labels match the SCCT categories [1, 2]. The percentage ranges below are those of the SCCT guideline and are assumed, not confirmed, for the CGPS reads.

Label	Grade	Diameter stenosis (SCCT)	Category
none	0	0 %	no plaque
minimal	1	1–24 %	non-obstructive
mild	2	25–49 %	non-obstructive
moderate	3	50–69 %	obstructive
severe	4	70–99 %	obstructive
occlusion	5	100 %	obstructive

2 Vessel-level summaries

Computed by the CGPS readers from the segment reads, for <v> ∈ {lad, cx, rca}.

Variable	BL	FU	Values	Role	Note
ct_cag_<v>_sis	✓	✓	Count of segments with plaque	check	Agrees with the segment reads in 95–97 % for CX and RCA but only 64–79 % for LAD; the recorded LAD score may include LM or IM.
ct_cag_<v>_sss	✓	✓	Sum of stenosis grades	check	As above.
ct_cag_<v>_plaque_type_score	✓	✓	Sum of plaque-type scores	not used	
ct_cag_<v>_uneval	✓	✓	Count of unevaluable segments	check	

Variable	BL	FU	Values	Role	Note
ct_cag_<v>_uneval_ contrast, _scanfield, _pm, _noise	–	✓	Count	check	Reasons for unevaluable segments at FU.
ct_cacs_<v> (lm, lad, cx, rca)	✓	✓	Agatston score	check, covariate	Calcium per vessel; secondary outcome ($0 \rightarrow >0$) and combined eligibility (trunk none and vessel calcium 0).

3 Patient-level summaries

Variable	BL	FU	Values	Role	Note
ct_cag_sis	✓	✓	Segment involvement score	covariate	Recorded as 0, not missing, for the 1 094 patients without BL segment reads.
ct_cag_sss	✓	✓	Segment stenosis score	covariate	As above.
ct_cag_plaque_type_ score	✓	✓	Sum of plaque-type scores	not used	
ct_cag_none … ct_cag_occlusion	✓	✓	Flags or counts per stenosis category	not used	Six variables (none, minimal, mild, moderate, severe, occlusion); all 0 for patients without BL reads.
ct_cag_dominance	✓	✓	Text	covariate	Needed to assign PDA and PLA.
ct_cag_stent, ct_cag_cabg	✓	✓	0/1	eligibility	Prior revascularisation; stent missing for ≈66 % at BL.
ct_cag_incomplete	✓	✓	Count	check	0 for the 1 022 read patients and 16–17 for the others at BL.
ct_cag_uneval	✓	✓	Count of unevaluable segments	check	
ct_cag_uneval_motion, _noise, _scanfield, _contrastdefect, _pm, _other	–	✓	Count	check	Reasons at FU; motion dominates.
ct_cag_eval_n	✓	–	Number of evaluable segments	check	Median 16.

Variable	BL	FU	Values	Role	Note
ct_cacs_sum	✓	✓	Total Agatston score	covariate, check	Available for 2 109 patients at both visits; calcium progression (Hokanson) is the most reliable outcome in the data.
ct_cacs_groups	✓	✓	Category 1–3	not used	
ct_cacs_<v>_eval, ct_cacs_eval_all	–	✓	0/1	check	Calcium evaluability at FU.

4 Clinical events and timing

Variable	BL	FU	Values	Role	Note
scandate	✓	✓	Date	covariate	Gives the follow-up time (median 10.0 years, 5th–95th percentile 9.5–15.3), used as log offset.
demo_hist_myocardial_infarction	✓	✓	0/1	not used	6 new events; missing for ≈51 % at BL.
demo_hist_coronary_revasc	✓	✓	0/1	not used	6 new events.
demo_hist_stroke_tia	✓	✓	0/1	not used	31 new; 119 report it at BL but not at FU.
hist_angina	✓	✓	0/1	not used	96 new.
demo_hist_afib, _heart_failure, _pad, _icd_pm, _cag	–	✓	0/1	not used	FU only; new cannot be separated from pre-existing.
ae_*	–	✓	Trial adverse events	not used	Related to the trial, not to coronary disease.

Hard clinical events (myocardial infarction, revascularisation, stent, bypass) total 10 over follow-up and cannot serve as an outcome.

5 Image quality

Variable	BL	FU	Values	Role	Note
scan_qualitative_quality	✓	–	0/1	check	Missing for 246.
bad_quality	✓	–	True/False	not used	False for every patient.
Image_Quality_reason_1, _2	✓	–	Free text	not used	Not reported by the script (text).
ct_angio_heart_done, ct_noncontrast_heart_done	–	✓	0/1	check	1 for every patient at FU; no BL counterpart, so BL contrast CT availability must be checked from the volumes.

6 Chosen design and derived outcomes

Element	Definition
Unit	LAD (plad, mlad, dlad), CX (pcx, mcx, dcx) and RCA (prca, mrca, drca) trunks; side branches excluded.
Eligibility	All trunk segments evaluable and graded none at BL; for LAD and CX the LM is also graded none.
Primary outcome	Any trunk segment graded minimal or worse at FU (binary). Unknown if no plaque is seen and a trunk segment is unevaluable at FU.
Secondary outcome	Worst FU grade in the trunk, ordinal: none, minimal, mild or worse.
Calcium check	Vessel calcium 0 at BL $\rightarrow$ >0 at FU (ct_cacs_lad, ct_cacs_cx, ct_cacs_rca).
Sensitivity	Eligibility additionally requiring vessel calcium 0 at BL; RCA additionally requiring LM none.
Follow-up time	log(years between scandate values) as offset with a complementary log-log link.

Open issues

- Whether the 1 094 patients without BL segment reads had a normal BL scan or no contrast CT; decides whether they can enter the design.
- The CGPS stenosis grading thresholds and segment boundary definitions (reading protocol).
- The meaning of the ct_cag_<s>_eval codes and of ct_cag_incomplete.

References

[1] J. Leipsic et al. SCCT guidelines for the interpretation and reporting of coronary CT angiography. *Journal of Cardiovascular Computed Tomography*, 8(5):342–358, 2014.

[2] R. C. Cury et al. CAD-RADS: Coronary Artery Disease – Reporting and Data System. *Journal of Cardiovascular Computed Tomography*, 10(4):269–281, 2016.